

SHETH L.U.J & M V COLLEGE

PRACTICAL NO 5 MOD 2

The screenshot displays the RStudio interface with the following components:

- Source Editor:** Contains R code for loading a dataset and performing a t-test.

```
1 library(dplyr)
2
3 df <- read.csv("medical_cost_prediction_dataset.csv")
4 colnames(df)
5
6 print("----- Independent Two Sample t-test: BMI by Gender -----")
7 t_test_two <- t.test(bmi ~ gender, data = df)
8 print(t_test_two)
9
```
- Environment:** Lists loaded objects: `sales_logic` (1000 obs. of 16 variables), `sales_text` (1000 obs. of 15 variables), `selected_cols` (200 obs. of 3 variables), `shopping_behav...` (3900 obs. of 18 variables), `sorted_by_volu...` (122 obs. of 7 variables), `split_matrix` (chr [1:1000, 1:2] "North" "West" "South..."), `starts_with_d` (200 obs. of 3 variables), `t_test_one` (List of 10), and `t_test_two` (List of 10).
- Console:** Shows the execution of the code and the output of the t-test.

```
> library(dplyr)
> df <- read.csv("medical_cost_prediction_dataset.csv")
> colnames(df)
[1] "age" "gender" "bmi"
[4] "smoker" "diabetes" "hypertension"
[7] "heart_disease" "asthma" "physical_activity_level"
[10] "daily_steps" "sleep_hours" "stress_level"
[13] "doctor_visits_per_year" "hospital_admissions" "medication_count"
[16] "insurance_type" "insurance_coverage_pct" "city_type"
[19] "previous_year_cost" "annual_medical_cost"
> print("----- Independent Two Sample t-test: BMI by Gender -----")
[1] "----- Independent Two Sample t-test: BMI by Gender -----"
> t_test_two <- t.test(bmi ~ gender, data = df)
> print(t_test_two)

Welch Two Sample t-test

data:  bmi by gender
t = 0.4793, df = 4996.8, p-value = 0.6317
alternative hypothesis: true difference in means between group Female and group Male is not equal to 0
95 percent confidence interval:
-0.2113909 0.3482059
sample estimates:
mean in group Female mean in group Male
26.00476 25.93636
```
- Files Panel:** Shows a list of files in the current directory, including `Downloads - ShortcutLink`, `flower_dataset - flower_dataset.csv`, `Iris - Iris.csv`, `medical_cost_prediction_dataset.csv`, `NetBeansProjects`, `penguins.csv`, `pushpal`, `pushpal prac`, `retail_store_sales.csv`, `sales_data.csv`, `sales_dataset.csv`, `sgfdu05m.ini`, `shopping_behavior_updated.csv`, `Updated Quality of Life Data.csv`, and `Virtual Machines`.

The Windows taskbar at the bottom shows the system clock as 23:57 on 03-01-2026, with a temperature of 24°C and various system icons.